Technical Appendix: Regime-Aware Experience Replay for Deep Reinforcement Learning in Portfolio Optimization

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A Experimental Configuration

This appendix provides complete hyperparameter specifications for all algorithms tested in our experiments. All experiments were conducted on a system with an NVIDIA RTX 5090 Laptop GPU (23.9GB VRAM), using the following software stack:

- **Core RL:** Python 3.12.10, PyTorch 2.10.0 (CUDA 12.8), Stable Baselines3, Gymnasium
- **Data processing:** pandas, NumPy, yfinance (market data retrieval)
- **Regime detection:** hmmlearn (Gaussian HMM), scikit-learn (preprocessing and clustering utilities)
- **Statistical analysis:** SciPy (Welch's t-tests, bootstrap confidence intervals)
- **Visualisation:** Matplotlib, seaborn

A.1 TD3 Hyperparameters

Twin Delayed DDPG (TD3) is our primary algorithm, achieving the best overall performance across all replay strategies.

Table 1: TD3 Hyperparameters

Parameter	**Value**	**Description**
Learning		
learning_rate	3×10^{-4}	Adam optimizer learning rate
batch_size	256	Mini-batch size for gradient updates
buffer_size	1,000,000	Replay buffer capacity
gamma (γ)	0.99	Discount factor
tau (τ)	0.005	Soft update coefficient
TD3-Specific		
policy_delay	2	Policy updates per critic update
target_policy_noise	0.2	Noise added to target actions
target_noise_clip	0.5	Clipping range for target noise
Training		
train_freq	4	Steps between gradient updates
gradient_steps	1	Gradient steps per update
learning_starts	10,000	Steps before training begins
total_timesteps	500,000	Total training steps
Network Architecture		
net_arch	[256, 256]	Hidden layer sizes (actor & critic)
activation_fn	ReLU	Activation function
Exploration		
action_noise	Normal	Gaussian exploration noise
noise_std	0.1	Standard deviation of action noise

A.2 SAC Hyperparameters

Soft Actor-Critic uses entropy regularization to encourage exploration. Our experiments revealed that SAC is incompatible with regime-aware replay due to entropy bias.

Table 2: SAC Hyperparameters

Parameter	Value	Description
Learning		
learning_rate	3×10^{-4}	Adam optimizer learning rate
batch_size	256	Mini-batch size
buffer_size	1,000,000	Replay buffer capacity
gamma (γ)	0.99	Discount factor
tau (τ)	0.005	Soft update coefficient
SAC-Specific		
ent_coef	auto	Automatic entropy tuning
target_entropy	auto	Automatic target entropy
Training		
train_freq	4	Steps between gradient updates
gradient_steps	1	Gradient steps per update
learning_starts	10,000	Steps before training begins
total_timesteps	500,000	Total training steps
Network Architecture		
net_arch	[256, 256]	Hidden layer sizes

A.3 PPO Hyperparameters

Proximal Policy Optimization is an on-policy algorithm that does not use experience replay. We make it regime-aware by augmenting observations with regime information.

Table 3: PPO Hyperparameters

Parameter	Value	Description
Learning		
learning_rate	3×10^{-4}	Adam optimizer learning rate
batch_size	64	Mini-batch size
n_steps	2,048	Steps to collect before update
n_epochs	10	Epochs per update
gamma (γ)	0.99	Discount factor
PPO-Specific		
gae_lambda	0.95	GAE parameter
clip_range	0.2	PPO clipping parameter
vf_coef	0.5	Value function loss coefficient
ent_coef	0.01	Entropy bonus coefficient
max_grad_norm	0.5	Gradient clipping threshold
Network Architecture		
net_arch	[256, 256]	Hidden layer sizes
Regime-Awareness		
add_regime_to_obs	True	Augment observations with regime

A.4 Rainbow DQN Hyperparameters

Rainbow combines multiple DQN improvements. We use a custom implementation with distributional RL, prioritized replay, and dueling networks.

Table 4: Rainbow DQN Hyperparameters

Parameter	**Value**	**Description**
Learning		
learning_rate	6.25×10^{-5}	Adam optimizer learning rate
batch_size	32	Mini-batch size
buffer_size	200,000	Replay buffer capacity
gamma (γ)	0.99	Discount factor
Distributional RL		
n_atoms	51	Number of atoms in distribution
v_min	-10.0	Minimum value support
v_max	10.0	Maximum value support
n_steps	3	Multi-step returns
Prioritized Replay		
per_alpha	0.6	Priority exponent
per_beta	0.4	Importance sampling exponent
Training		
target_update_freq	8,000	Target network update frequency
learning_starts	20,000	Steps before training begins
train_freq	4	Steps between gradient updates
total_timesteps	500,000	Total training steps
Network Architecture		
hidden_dims	[256, 256]	Hidden layer sizes
noisy_sigma	0.5	NoisyNet initial σ

A.5 Environment Configuration

Table 5: Trading Environment Configuration

Parameter	Value	Description
Assets		
universe	SPY, QQQ, TLT, GLD, BIL	5-asset portfolio
initial_capital	$1,000,000	Starting portfolio value
Costs		
transaction_cost	0.001 (10 bps)	Per-trade cost
Reward		
reward_scaling	1.0	Reward multiplier
λ_{cost}	1.0	Transaction cost penalty coefficient
λ_{risk}	0.1	Drawdown penalty coefficient
DD_threshold	0.10	Drawdown threshold before penalty
Data Splits		
train_period	2010-01-01 to 2019-12-31	Training data
val_period	2020-01-01 to 2021-12-31	Validation data
test_period	2022-01-01 to 2023-12-31	Test data

B Research Hypotheses

Table 6 summarises the six research hypotheses that guided our experimental programme. Each hypothesis is mapped to the experiment(s) designed to test it, along with the outcome based on our results.

Table 6: Research hypotheses, associated experiments, and outcomes.

ID	Hypothesis	Test	Outcome
H1	Regime-aware replay improves risk-adjusted returns vs standard uniform replay	E1	**Not supported.** Standard replay yields the best Sharpe (+0.632); no regime-aware strategy significantly outperforms it ($p > 0.05$)
H2	Balanced sampling reduces performance variance across market regimes	E1, E2	**Not supported.** Balanced sampling does not reduce cross-regime variance and actively harms SAC ($d = -2.95$)
H3	Inverse-frequency sampling improves bear market performance	E2	**Not supported.** TD3's bear market advantage stems from the algorithm, not the sampling strategy
H4	Current-focused sampling improves short-term adaptability	E2	**Partially supported.** TD3 current-focused achieves the highest bear regime Sharpe (+1.07) but ranks second overall
H5	Benefits are consistent across all four algorithms	E1	**Rejected.** SAC is fundamentally incompatible with non-uniform replay; effects are algorithm-dependent
H6	Robust to moderate noise ($\leq$20%) in regime detection	E3	**Supported.** No significant degradation up to 30% noise ($p > 0.05$)

C Full Results Tables

This appendix provides complete result tables for all seven experiments. The primary metric is the annualized Sharpe ratio computed on the 2022–2023 test period. Secondary metrics include the Sortino ratio (return per unit of downside risk), Calmar ratio (annualized return divided by maximum drawdown), and cumulative return. MDD denotes maximum drawdown. Val–Test Gap is the difference between validation and test Sharpe ratios; larger gaps indicate greater overfitting.

C.1 E1: Replay Strategy Comparison (100 Runs)

Table 7: Complete E1 results: 4 algorithms × 5 strategies × 5 seeds. Sorted by algorithm, then strategy.

Algorithm	Strategy	Sharpe	± Std	Sortino	Ret. (%)	Calmar	Val Sharpe	MDD (%)	Gap
PPO	adaptive	+0.281	0.185	+0.442	+8.1	+0.226	1.359	19.4	1.078
PPO	balanced	+0.168	0.205	+0.261	+5.1	+0.133	1.503	21.1	1.335
PPO	current_focused	+0.325	0.293	+0.509	+8.3	+0.314	1.411	16.9	1.086
PPO	inverse_freq	+0.212	0.236	+0.330	+5.8	+0.159	1.876	20.8	1.664
PPO	standard	+0.289	0.288	+0.455	+7.6	+0.245	1.313	17.6	1.024
Rainbow	adaptive	−0.007	0.057	−0.010	−0.2	−0.003	1.321	20.5	1.328
Rainbow	balanced	+0.060	0.046	+0.093	+1.3	+0.035	1.345	19.1	1.285
Rainbow	current_focused	−0.024	0.073	−0.037	−0.5	−0.014	1.408	20.7	1.432
Rainbow	inverse_freq	+0.091	0.126	+0.145	+2.2	+0.059	1.349	20.0	1.258
Rainbow	standard	+0.062	0.080	+0.098	+1.5	+0.039	1.462	20.1	1.400
SAC	adaptive	+0.171	0.213	+0.268	+4.1	+0.113	1.165	18.6	0.994
SAC	balanced	−0.014	0.173	−0.012	−0.3	−0.006	1.337	21.7	1.351
SAC	current_focused	+0.084	0.205	+0.127	+2.0	+0.057	1.213	20.9	1.129
SAC	inverse_freq	+0.134	0.164	+0.205	+3.3	+0.091	1.341	19.1	1.207
SAC	standard	+0.458	0.146	+0.728	+13.3	+0.374	1.702	17.1	1.244
TD3	adaptive	+0.501	0.127	+0.795	+16.1	+0.496	1.444	16.3	0.943
TD3	balanced	+0.438	0.162	+0.661	+14.4	+0.393	1.415	18.6	0.977
TD3	current_focused	+0.620	0.333	+0.999	+19.8	+0.654	1.725	17.3	1.105
TD3	inverse_freq	+0.254	0.188	+0.403	+7.9	+0.210	1.608	20.0	1.354
TD3	standard	**+0.632**	0.166	+0.998	+19.7	+0.645	1.537	15.2	**0.905**

C.2 E3: Robustness to Noisy Regime Detection (36 Runs)

TD3 only, with regime label noise at 0%, 10%, 20%, and 30%.

Table 8: E3 results: TD3 with varying regime detection noise levels (3 seeds per config).

Noise	Strategy	Sharpe	± Std	Sortino	Ret. (%)	Val Sharpe	Gap
0%	standard	+0.540	0.123	+0.844	+18.6	1.850	1.310
0%	balanced	+0.193	0.007	+0.295	+6.4	2.061	1.868
0%	inverse_freq	+0.485	0.368	+0.787	+15.2	1.315	0.830
10%	standard	+0.620	0.111	+0.969	+21.2	1.529	0.909
10%	balanced	+0.734	0.098	+1.176	+23.8	1.625	0.891
10%	inverse_freq	+0.288	0.223	+0.454	+8.7	2.025	1.737
20%	standard	+0.640	0.074	+1.023	+20.8	1.417	0.777
20%	balanced	+0.585	0.226	+0.909	+19.3	1.344	0.759
20%	inverse_freq	+0.403	0.213	+0.649	+13.1	1.445	1.042
30%	standard	+0.652	0.085	+1.076	+20.7	1.752	1.100
30%	balanced	+0.312	0.260	+0.489	+10.5	1.448	1.136
30%	inverse_freq	+0.547	0.493	+0.902	+18.3	1.770	1.223

C.3 E4: Feature Ablation and E5: Detection Method

Table 9: E4 feature ablation (20 runs: PPO and SAC, 5 seeds each) and E5 detection method comparison (12 runs: SAC, 3 seeds each).

Exp	Config 1	Config 2	Sharpe	± Std	Sortino	Ret. (%)	MDD (%)	Gap
E4: Feature Ablation								
E4	PPO	full	+0.155	0.187	+0.245	+4.2	21.2	1.335
E4	PPO	reduced	+0.179	0.106	+0.269	+5.6	23.1	1.001
E4	SAC	full	−0.052	0.322	−0.074	−1.2	23.2	1.278
E4	SAC	reduced	+0.074	0.162	+0.121	+1.9	19.1	1.242
E5: Detection Method (SAC only)								
E5	HMM	standard	+0.644	0.132	+1.058	+19.3	14.8	0.986
E5	HMM	balanced	−0.036	0.189	−0.054	−0.7	19.8	1.396
E5	Rule-based	standard	+0.617	0.087	+0.995	+18.3	13.7	1.164
E5	Rule-based	balanced	+0.001	0.079	+0.005	+0.0	21.8	1.234

C.4 E6: Sample Efficiency and E7: Timestep Ablation

Table 10: E6 sample efficiency (30 runs: TD3 and SAC at 1M timesteps, 5 seeds) and E7 timestep ablation (30 runs: TD3 only, 3 seeds).

Exp	Config	Strategy	Sharpe	± Std	Sortino	Ret. (%)	Val Sharpe	Gap
E6: Sample Efficiency (1M timesteps)								
E6	TD3	inverse_freq	**+0.695**	0.315	+1.136	+22.3	1.658	0.963
E6	TD3	balanced	+0.567	0.222	+0.913	+18.6	1.764	1.197
E6	TD3	standard	+0.432	0.425	+0.686	+13.8	1.525	1.093
E6	SAC	balanced	+0.338	0.291	+0.549	+9.1	1.255	0.917
E6	SAC	standard	+0.335	0.102	+0.543	+10.0	1.524	1.189
E6	SAC	inverse_freq	+0.256	0.361	+0.415	+6.9	1.298	1.042
E7: Timestep Ablation (TD3 only)								
E7	100k	adaptive	+0.624	0.160	+0.974	+19.3	1.613	0.990
E7	100k	standard	+0.550	0.194	+0.886	+17.1	1.655	1.105
E7	200k	adaptive	+0.681	0.054	+1.065	+22.3	1.361	0.680
E7	200k	standard	+0.373	0.312	+0.593	+11.8	1.193	0.820
E7	300k	adaptive	+0.382	0.289	+0.607	+11.7	1.499	1.117
E7	300k	standard	+0.356	0.122	+0.576	+11.4	1.457	1.101
E7	400k	adaptive	+0.365	0.138	+0.557	+11.8	1.680	1.315
E7	400k	standard	**+0.815**	0.151	+1.312	+25.7	1.422	**0.607**
E7	500k	adaptive	+0.738	0.195	+1.152	+24.6	1.586	0.848
E7	500k	standard	+0.649	0.175	+0.989	+21.3	1.528	0.879

D Statistical Analysis Details

This appendix provides the complete statistical analysis supporting the results in Section 5 of the main report.

Methodology. All comparisons use Welch's t-test (unequal variances assumed) with each random seed treated as an independent observation ($n = 5$ per group for E1). We apply Bonferroni correction for multiple comparisons: with 16 pairwise tests in E1 (4 algorithms $\times$ 4 non-standard strategies vs standard), the corrected significance threshold is $\alpha = 0.05/16 = 0.003125$. Effect sizes are reported as Cohen's d, interpreted as: negligible (< 0.2), small (0.2–0.5), medium (0.5–0.8), and large (> 0.8).

D.1 E1 Pairwise t-Tests: Strategy vs Standard

Table 11: Pairwise t-tests comparing each regime-aware strategy against standard (uniform) replay within each algorithm. Bonferroni-corrected $\alpha = 0.003125$.

Algorithm	**Strategy**	**t-stat**	**p-value**	**Cohen's d**	**Bonf.**
PPO	balanced	−0.76	0.469	−0.48	No
PPO	current_focused	+0.20	0.849	+0.12	No
PPO	inverse_freq	−0.46	0.659	−0.29	No
PPO	adaptive	−0.05	0.965	−0.03	No
Rainbow	balanced	−0.06	0.954	−0.04	No
Rainbow	current_focused	−1.78	0.112	−1.13	No
Rainbow	inverse_freq	+0.42	0.685	+0.27	No
Rainbow	adaptive	−1.58	0.157	−1.00	No
SAC	balanced	−4.66	**0.00175**	−2.95	**Yes**
SAC	current_focused	−3.32	0.012	−2.10	No
SAC	inverse_freq	−3.30	0.011	−2.09	No
SAC	adaptive	−2.49	0.041	−1.57	No
TD3	balanced	−1.86	0.099	−1.18	No
TD3	current_focused	−0.07	0.948	−0.04	No
TD3	inverse_freq	−3.37	0.010	−2.13	No
TD3	adaptive	−1.40	0.203	−0.88	No

All values are from pairwise Welch's t-tests on per-seed test Sharpe ratios ($n = 5$ per group). SAC balanced is the only Bonferroni-significant comparison ($p = 0.00175$, $d = -2.95$). SAC current-focused, inverse-frequency, and adaptive all show raw-significant declines ($p = 0.012$, 0.011, and 0.041 respectively, $|d| > 1.5$) that do not survive Bonferroni correction. Critically, all four SAC non-standard strategies show large negative effect sizes ($|d| > 1.0$), indicating a consistent pattern of degradation rather than random variation. No significant effects are observed for PPO after correction. TD3 inverse-frequency shows a raw-significant decline ($p = 0.010$, $d = -2.13$) that does not survive Bonferroni correction. TD3 adaptive also shows a large negative effect ($d = -0.88$, $p = 0.203$), consistent with its lower post-correction mean (+0.501 vs standard +0.632).

D.2 E2 Bear Market Statistical Tests

Table 12: Statistical tests for regime-level performance claims (E2).

Comparison	**p-value**	**Cohen's d**	**Diff**	**Sig.**
TD3 vs PPO (bear)	2.4×10^{-5}	+1.33	+0.433	Yes
TD3 vs SAC (bear)	4.1×10^{-7}	+1.66	+0.586	Yes
TD3 vs Rainbow (bear)	7.6×10^{-11}	+2.93	+0.758	Yes
SAC std vs bal (bear)	0.032	+1.68	+0.551	No
SAC std vs bal (bull)	0.680	+0.27	+0.136	No
SAC std vs bal (sideways)	0.011	+2.12	+0.928	No
TD3 adaptive vs std (bear)	0.930	−0.06	−0.020	No

TD3's bear market superiority over all other algorithms is highly significant ($p < 0.001$ for all three comparisons), with large to very large effect sizes. SAC standard vs balanced shows large effects in bear and sideways regimes but does not reach significance after correction due to smaller sample sizes within each regime.

D.3 Win Rate Analysis

Win rates measure the proportion of seeds where a regime-aware strategy outperforms standard replay for the same algorithm (threshold: Sharpe difference > 0.01).

Table 13: Win rates: regime-aware strategy vs standard replay, by seed (5 seeds per comparison). A "win" requires a Sharpe difference > 0.01; differences ≤ 0.01 are ties.

Algorithm	Strategy	Wins	Losses	Ties	Win Rate
PPO	balanced	2	3	0	40%
PPO	current_focused	2	3	0	40%
PPO	inverse_frequency	1	4	0	20%
PPO	adaptive	1	4	0	20%
Rainbow	balanced	2	3	0	40%
Rainbow	current_focused	0	5	0	0%
Rainbow	inverse_frequency	3	2	0	60%
Rainbow	adaptive	1	3	1	20%
SAC	balanced	0	5	0	0%
SAC	current_focused	0	5	0	0%
SAC	inverse_frequency	0	5	0	0%
SAC	adaptive	0	5	0	0%
TD3	balanced	2	3	0	40%
TD3	current_focused	3	2	0	60%
TD3	inverse_frequency	0	5	0	0%
TD3	adaptive	1	4	0	20%

SAC's 0% win rate across all four non-standard strategies (losing on all 5 seeds in each case) provides additional evidence for the incompatibility finding. TD3 adaptive achieves a 20% win rate (1 of 5 seeds), consistent with its lower mean Sharpe (+0.501) compared to standard (+0.632) after the EMA decay bug fix. TD3 inverse-frequency shows a 0% win rate, mirroring its near-significant decline in Table 11.

E All Figures

This appendix collects all experimental figures. Figures referenced in the main report are reproduced here for completeness alongside additional visualizations.

E.1 E1: Replay Strategy Comparison

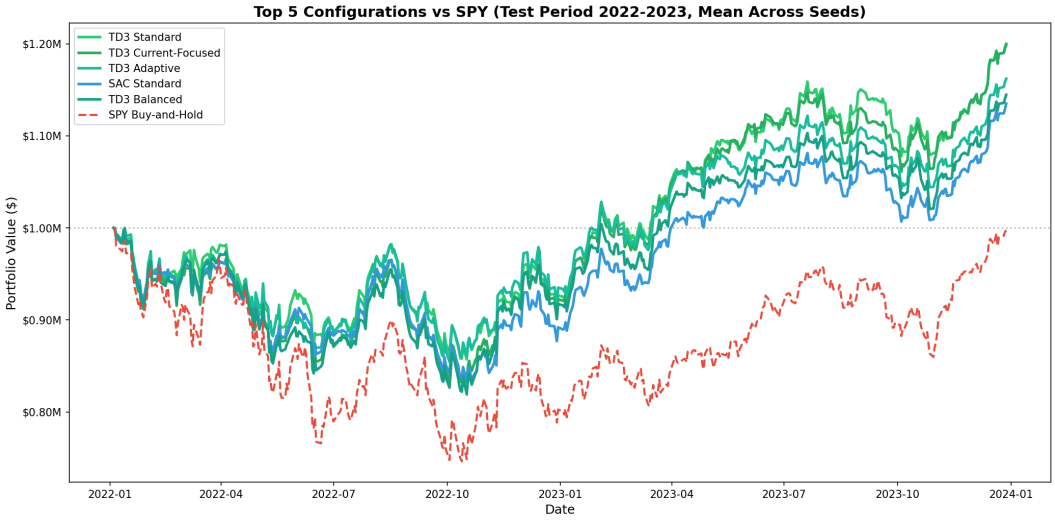


Figure 1: Equity curves for the top 5 algorithm–strategy configurations versus the SPY benchmark on the 2022–2023 test period (mean across 5 seeds). Starting from \$1M, all five configurations outperform SPY. TD3 standard achieves the best Sharpe ratio of +0.632 with a maximum drawdown of 15.2% compared to SPY's 25.3%.

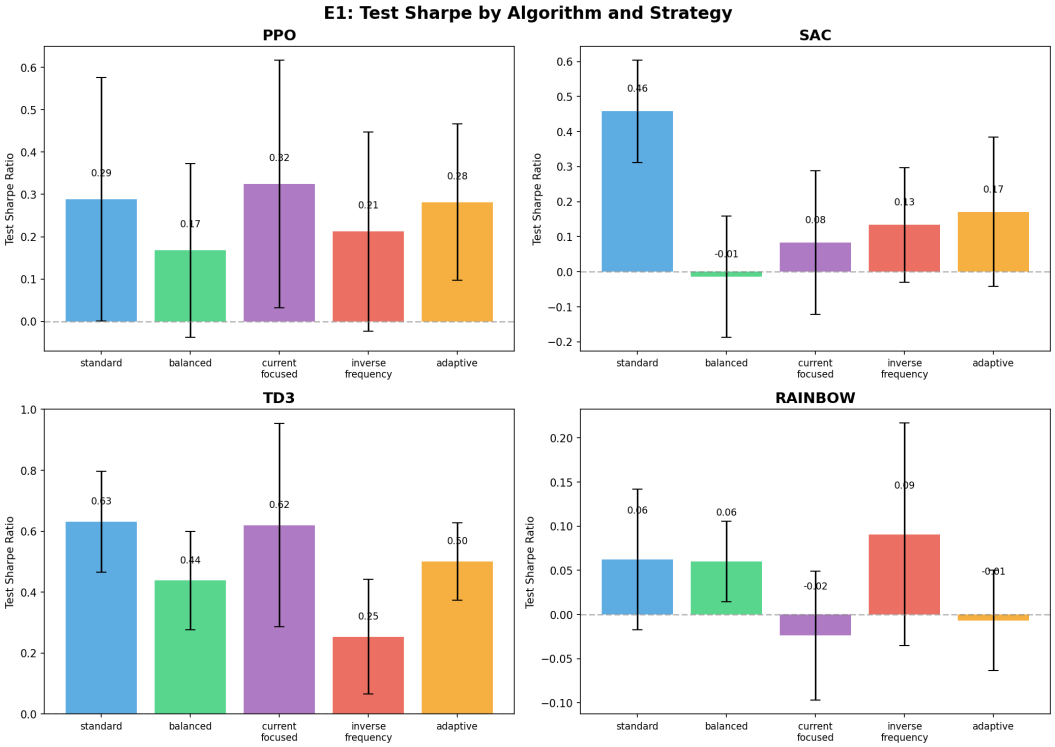


Figure 2: Test Sharpe ratios across all 20 algorithm–strategy combinations (E1, 5 seeds per bar, error bars show ±1 standard deviation). TD3 performs consistently well across all strategies. SAC collapses with any non-standard replay. Rainbow DQN shows uniformly low test performance despite high validation Sharpe ratios, indicating severe overfitting.

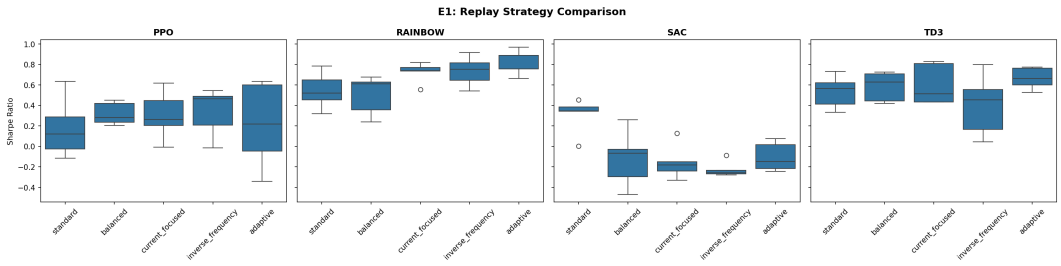


Figure 3: Box plots of test Sharpe ratios by algorithm (E1, all strategies pooled). TD3 has the highest median and smallest interquartile range. SAC shows a bimodal distribution driven by the standard-vs-regime-aware split. Rainbow DQN clusters near zero with minimal variance.

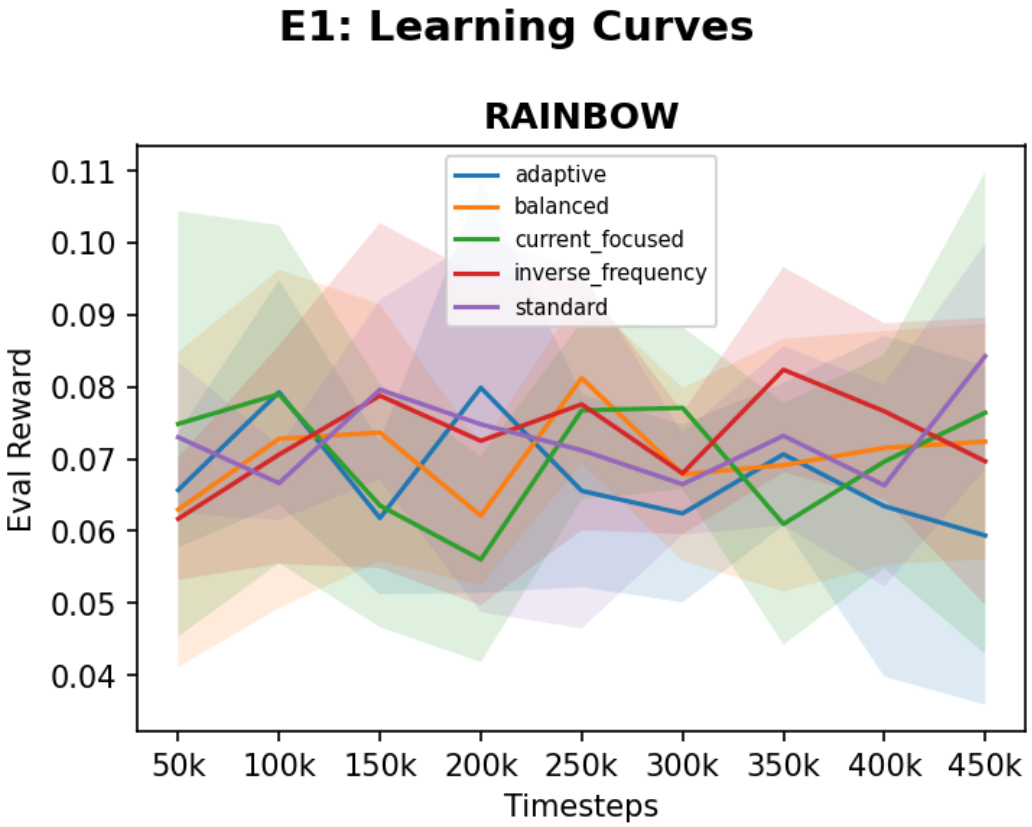


Figure 4: Training learning curves for each algorithm (E1, averaged across strategies and seeds). TD3 and SAC converge within 300k timesteps, while PPO shows more gradual improvement. Rainbow DQN converges quickly on the training set but this does not translate to test performance.

E.2 E2: Regime-Level Analysis

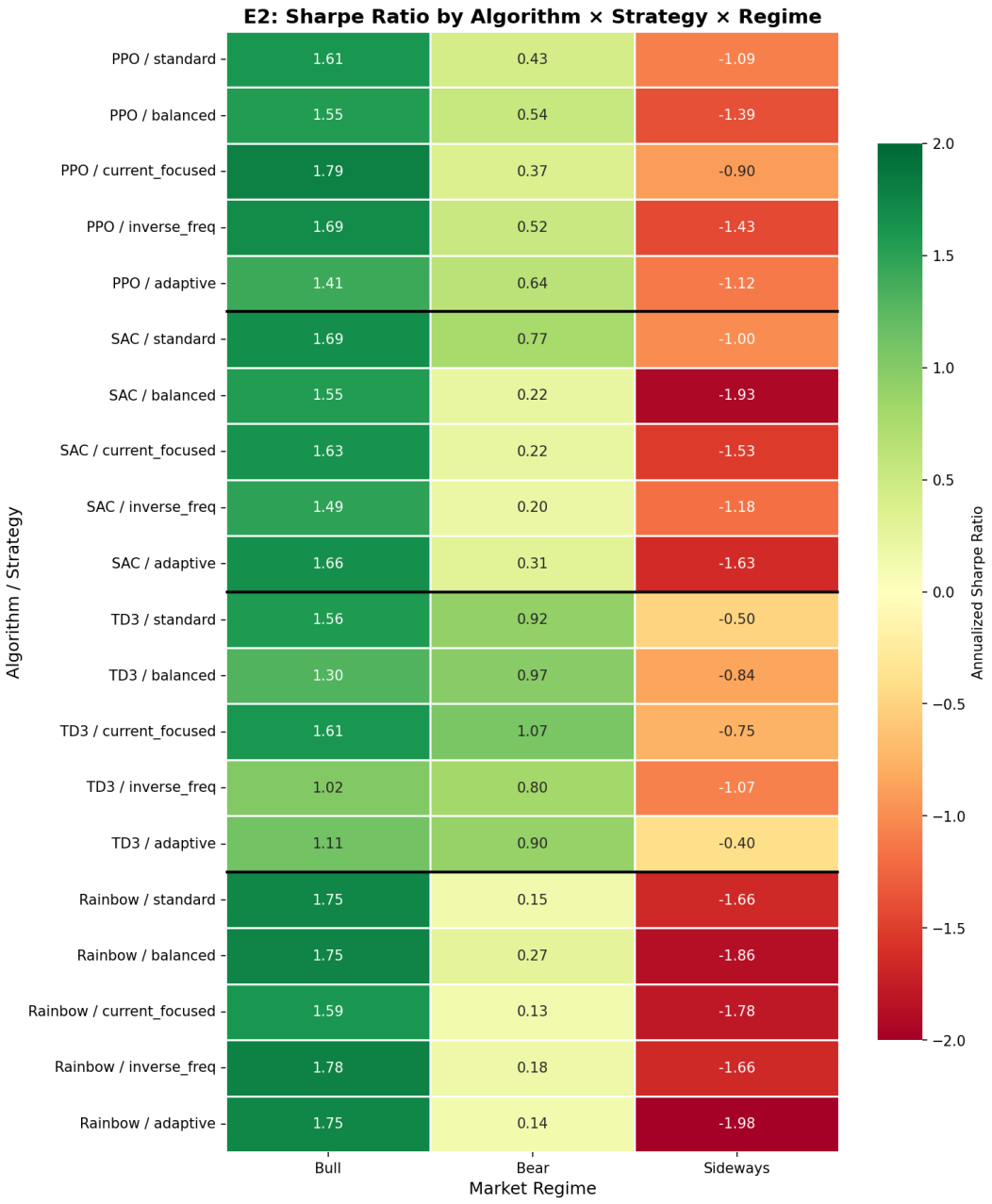


Figure 5: Annualized Sharpe ratio by algorithm, strategy, and market regime (E2, mean across seeds and episodes). TD3 dominates bear markets (+0.931 algorithm mean) while all algorithms profit in bull markets (+1.564 aggregate) and struggle in sideways conditions (−1.286 aggregate). TD3 current-focused achieves the highest bear regime Sharpe (+1.07). SAC standard performs well in bear markets (+0.77) but all non-standard SAC strategies collapse.

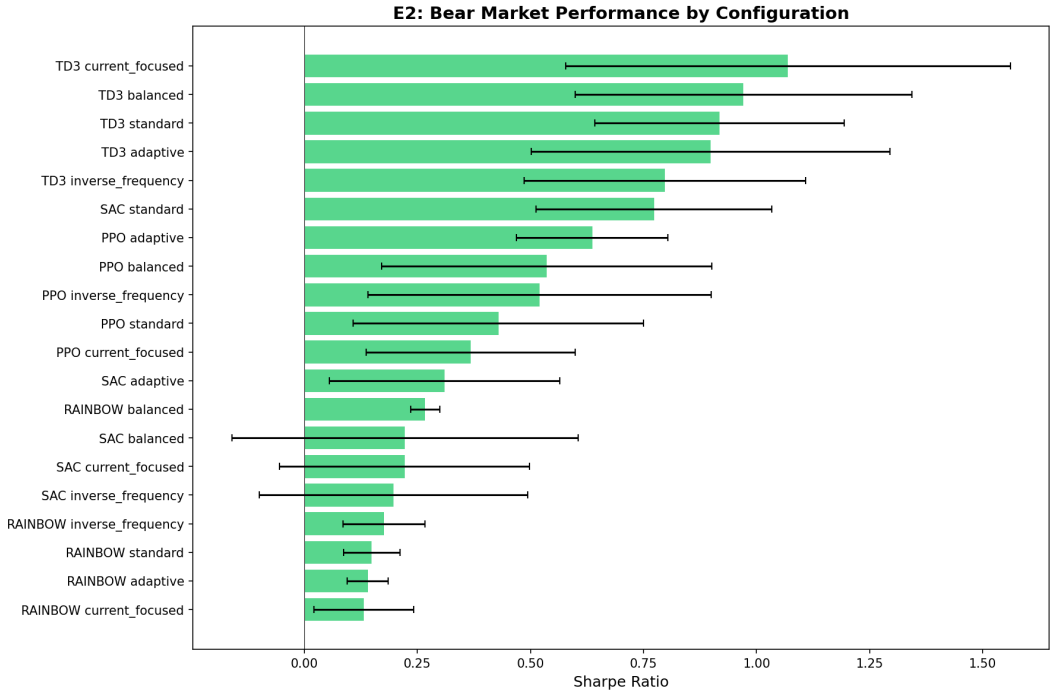


Figure 6: Bear market Sharpe ratio rankings by algorithm–strategy configuration (E2). TD3 configurations dominate the top positions, with TD3 current-focused achieving +1.069. The gap between TD3 and other algorithms is statistically significant ($p < 0.001$ for all pairwise comparisons).

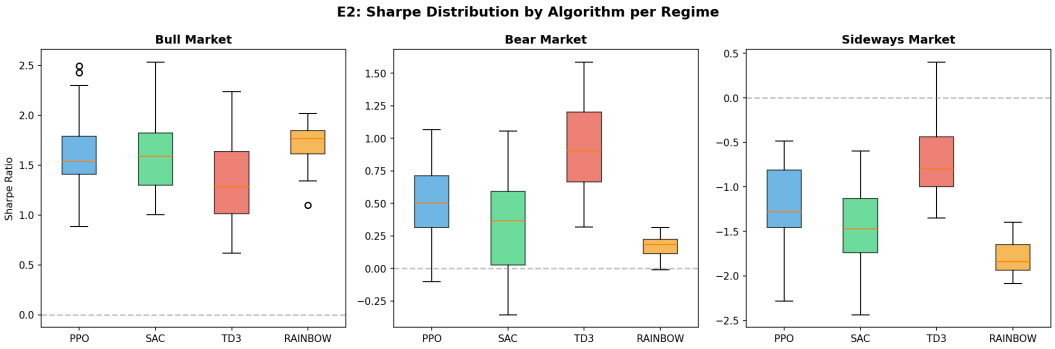


Figure 7: Distribution of Sharpe ratios by market regime across all 100 E1 runs. Bull markets show consistently positive Sharpe ratios with moderate variance. Bear markets display wider dispersion but remain predominantly positive. Sideways markets are universally challenging, with all configurations producing negative median Sharpe ratios.

E.3 E3–E7: Secondary Experiments

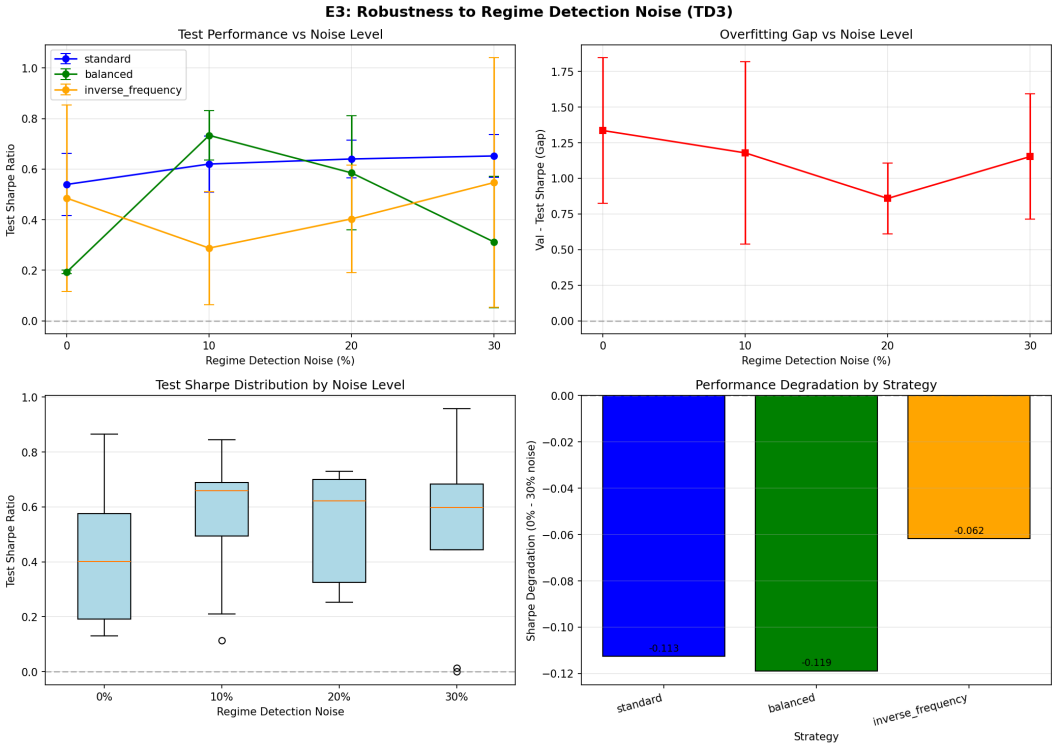


Figure 8: E3 robustness to noisy regime detection (TD3, 36 runs). Four panels show the effect of increasing noise (0–30%) on test Sharpe, validation–test gap, maximum drawdown, and strategy comparison. Standard replay is the most robust strategy, maintaining positive Sharpe ratios even at 30% label noise. Notably, moderate noise (10–20%) sometimes improves performance, suggesting a regularization effect.

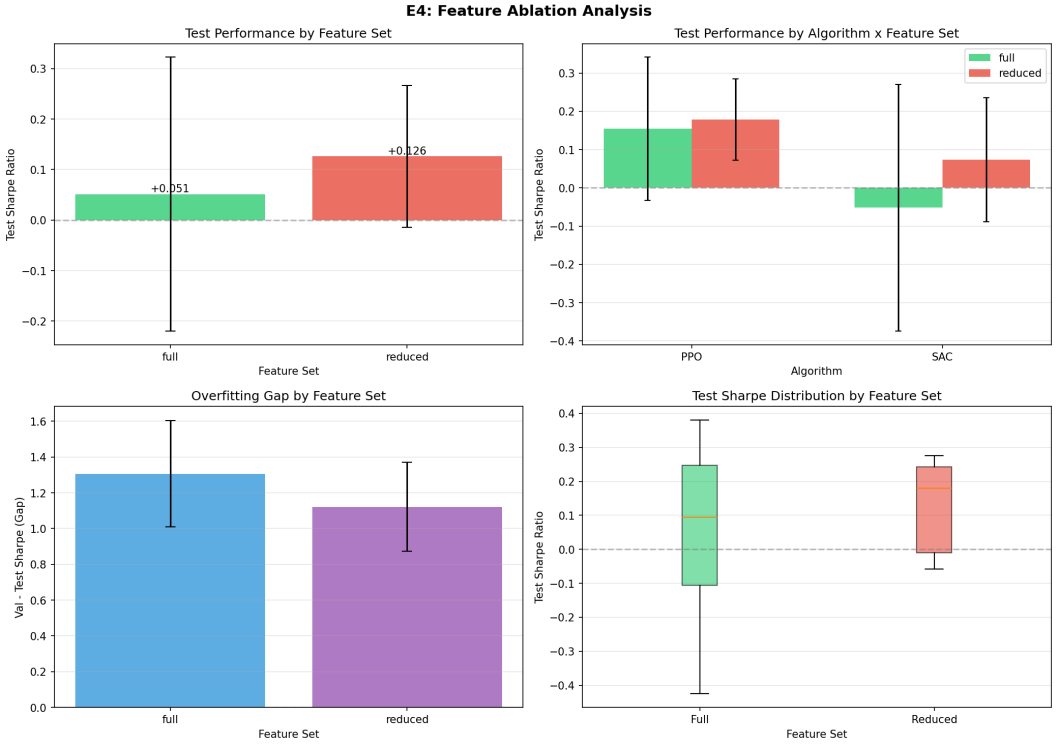


Figure 9: E4 feature ablation (20 runs: PPO and SAC). Comparison of full (92 features) versus reduced feature sets. Reduced features achieve comparable or slightly better test Sharpe ratios with consistently smaller validation–test gaps, suggesting that the full feature set contributes to overfitting without improving generalization.

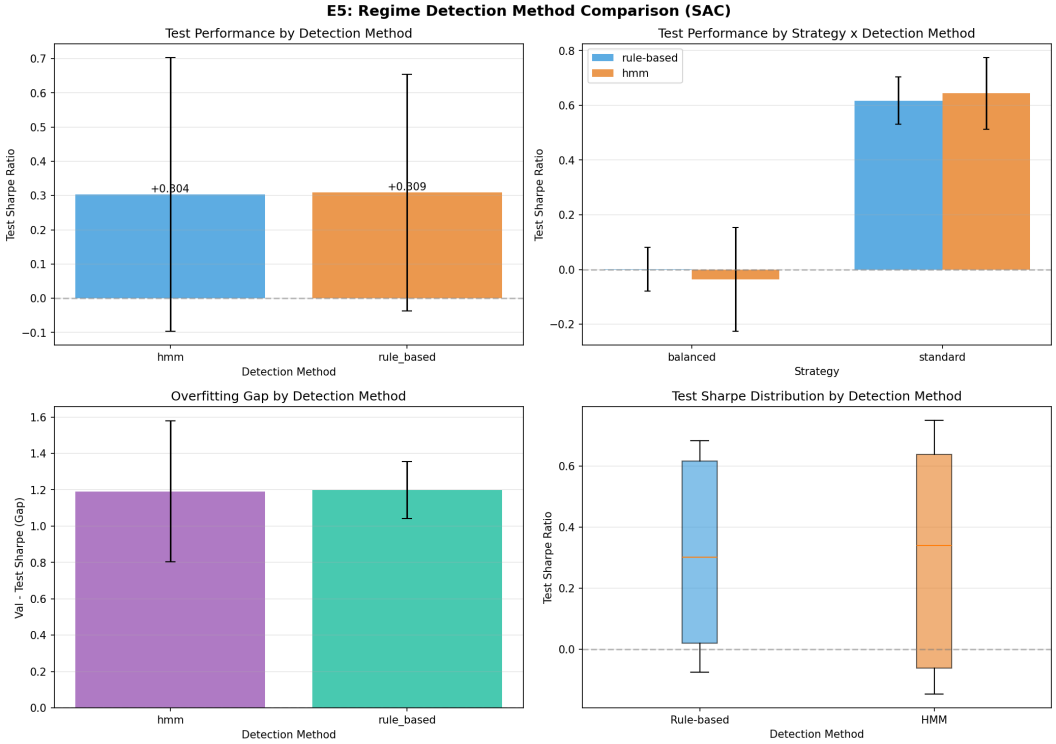


Figure 10: E5 regime detection method comparison (12 runs: SAC). HMM-based and rule-based detectors produce nearly identical results when using standard replay (HMM: +0.644, rule-based: +0.617). With balanced replay, both methods degrade, confirming that the SAC incompatibility is strategy-driven rather than detection-method-driven.

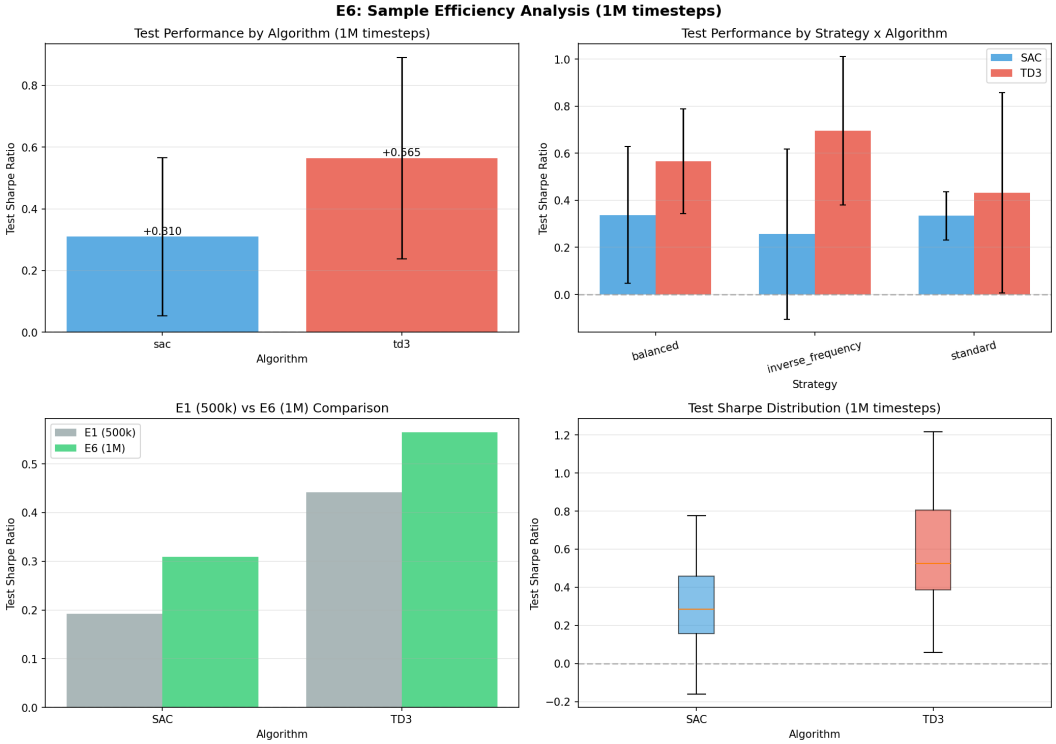


Figure 11: E6 sample efficiency at 1M timesteps (30 runs: TD3 and SAC). TD3 achieves higher Sharpe ratios than SAC across all replay strategies, with TD3 inverse-frequency reaching +0.695. Extended training beyond 500k timesteps provides diminishing returns for both algorithms.

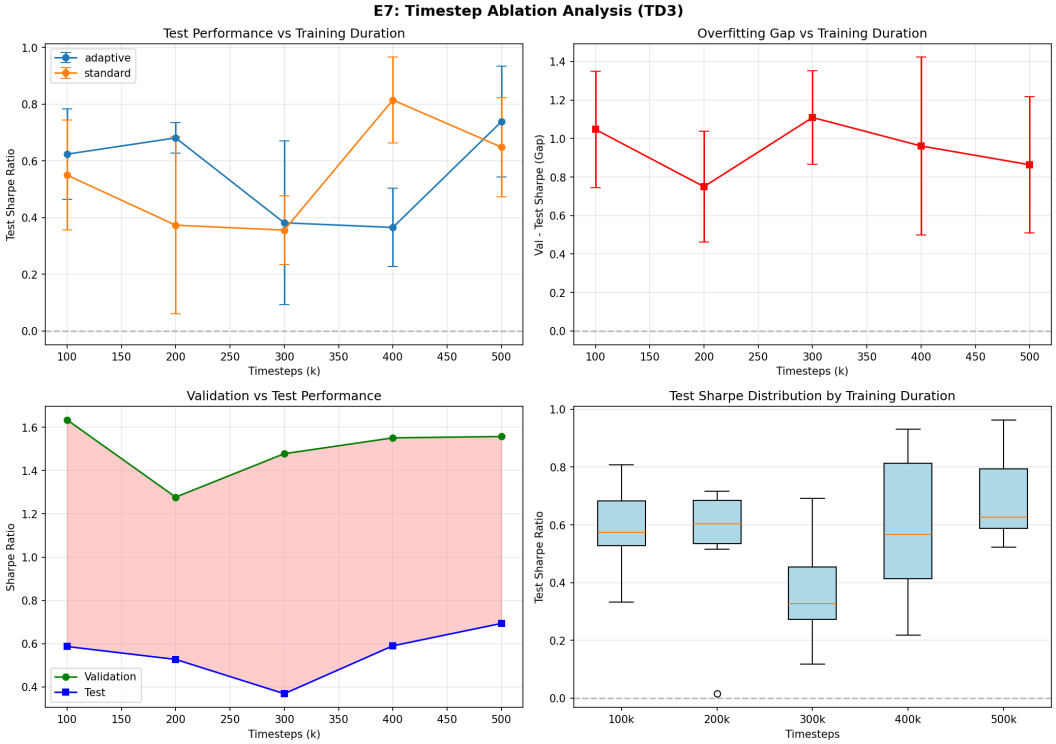


Figure 12: E7 timestep ablation for TD3 (30 runs). Test Sharpe and validation–test gap as a function of training duration (100k–500k timesteps). The optimal point is 400k timesteps with standard replay, which achieves both the highest test Sharpe (+0.815) and the lowest validation–test gap (0.607). Training beyond 400k leads to mild overfitting, as evidenced by increasing gaps without corresponding test improvements.

E.4 Context and Baselines

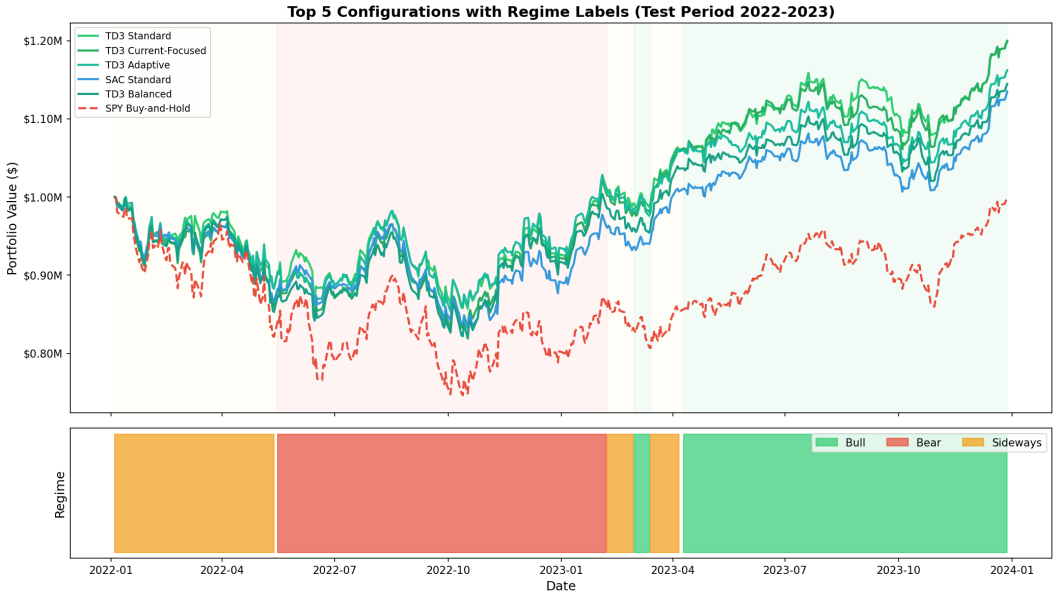


Figure 13: Portfolio value trajectories for the top 5 configurations with market regime shading (bull in green, bear in red, sideways in yellow) over the 2022–2023 test period. The regime detector identifies the 2022 bear market and the 2023 recovery, which the RL agents exploit by adjusting portfolio allocation.

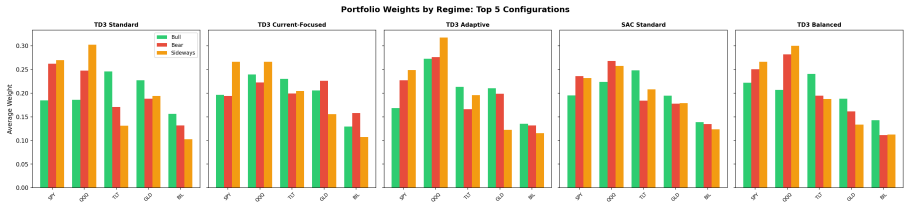


Figure 14: Mean portfolio weights by market regime for the top 5 configurations. Counter-intuitively, agents allocate more to risk assets (SPY+QQQ) during bear markets than during bull markets, suggesting exploitation of mean-reversion opportunities within bear-market rallies.

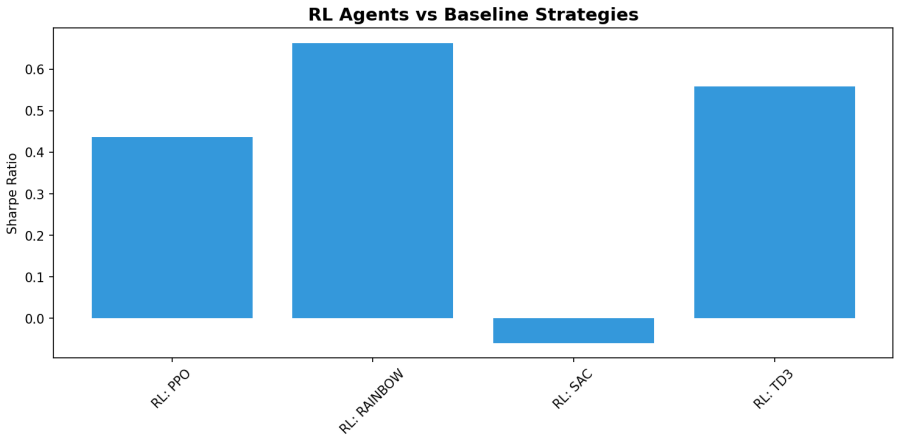


Figure 15: Test Sharpe ratios for passive baselines and top RL configurations. All four baselines produce negative or near-zero Sharpe ratios during the 2022–2023 period, while RL agents—particularly TD3—achieve substantially positive risk-adjusted returns.

F Implementation Details

This appendix provides key code excerpts from the regime-aware replay buffer, the core novel component of our system.

F.1 Regime-Aware Replay Buffer: Sampling Strategies

The `RegimeAwareReplayBuffer` partitions experiences by regime and samples according to a configurable strategy. The `sample()` method dispatches to strategy-specific implementations:

Listing 1: Main sampling dispatcher

```
def sample(self, batch_size, current_regime=None):
    if self.strategy == SamplingStrategy.BALANCED:
        return self._sample_balanced(batch_size)
    elif self.strategy == SamplingStrategy.CURRENT_FOCUSED:
        return self._sample_current_focused(batch_size, current_regime)
    elif self.strategy == SamplingStrategy.INVERSE_FREQUENCY:
        return self._sample_inverse_frequency(batch_size)
    elif self.strategy == SamplingStrategy.ADAPTIVE:
        return self._sample_adaptive(batch_size)
```

F.1.1 Balanced Sampling

Draws equal numbers from each regime buffer, with fallback redistribution when a regime has fewer samples than requested:

Listing 2: Balanced sampling implementation

```
def _sample_balanced(self, batch_size):
    available = {r: len(buf) for r, buf in self._regime_buffers.items()
        }
    base_per_regime = batch_size // len(MarketRegime)  # 3 regimes

    samples_per_regime = {}
    remaining = batch_size
    for regime in MarketRegime:
        n = min(base_per_regime, available[regime])
        samples_per_regime[regime] = n
        remaining -= n

    # Distribute remaining to regimes with extras
    for regime in shuffled(MarketRegime):
        if remaining <= 0: break
        extra = min(remaining, available[regime] - samples_per_regime[
            regime])
        samples_per_regime[regime] += extra
        remaining -= extra

    # Sample indices from each regime buffer
    for regime, n in samples_per_regime.items():
        indices = np.random.choice(len(buf), n, replace=False)
        experiences.extend([buf[i] for i in indices])
```

F.1.2 Inverse-Frequency Sampling

Oversamples rare regimes by weighting inversely to buffer size:

Listing 3: Inverse-frequency sampling

```
def _sample_inverse_frequency(self, batch_size):
    counts = {r: len(buf) for r, buf in self._regime_buffers.items()}
    weights = {r: 1.0 / (count + 1) for r, count in counts.items()}
    total_weight = sum(weights.values())
    proportions = {r: w / total_weight for r, w in weights.items()}
    # Sample batch_size * proportions[r] from each regime
```

F.1.3 Adaptive Sampling with TD Error Tracking

Samples proportionally to mean TD error per regime, using EMA decay:

Listing 4: Adaptive sampling and TD error update

```
def _sample_adaptive(self, batch_size):
    mean_td = {r: self._td_errors_sum[r] / max(self._td_errors_count[r
        ], 1)
                for r in MarketRegime}
    total_td = sum(mean_td.values())
    if total_td <= 0:
        return self._sample_balanced(batch_size)  # fallback
    proportions = {r: td / total_td for r, td in mean_td.items()}
    # Sample batch_size * proportions[r] from each regime

def update_td_errors(self, regime, td_errors):
    """Update TD error stats using EMA (decay=0.99)."""
    decay = self._td_ema_decay  # 0.99
```

```
        self._td_errors_sum[regime] = (
            decay * self._td_errors_sum[regime] + np.sum(np.abs(td_errors))
        )
        self._td_errors_count[regime] = (
            decay * self._td_errors_count[regime] + len(td_errors)
        )
```

G Bug Fix Documentation

During development, we identified and fixed eleven bugs that significantly impacted experimental results. Nine were corrected before initial execution; two additional bugs (#10, #11) were discovered post-execution, and affected experiments were re-run. This documentation demonstrates the engineering rigor applied to ensure correct metric calculation.

Table 14: Bug Fixes Applied

#	Bug	Impact	File
1	Episode length (val/test)	Incorrect episode termination	experiment_runner.py
2	Rainbow regime_detector	Missing regime detector	rainbow_agent.py
3	agent_factory regime_detector	Missing parameter passing	agent_factory.py
4	E3 SAC→TD3	Wrong algorithm for robustness	run_config.py
5	Transaction cost double-count	Costs subtracted twice	trading_env.py
6	Cumulative return compounding	Incorrect return calculation	experiment_runner.py
7	Test/Val date mismatch	Wrong date ranges	experiment_runner.py
8	N_EVAL_EPISODES mismatch	Inconsistent episode counts	experiment_runner.py
9	Adaptive TD error accumulation	TD errors accumulated without decay	regime_aware_buffer.py
10	Bug #9 incomplete: `add()`	`add()` still accumulated TD errors without EMA	regime_aware_buffer.py
11	Risk parity look-ahead bias	`iloc[-60:]` used future data	compute_baselines.py

The most impactful bugs were #5 (transaction cost double-counting), which inflated cost penalties by subtracting transaction costs in both the reward calculation and the portfolio value update, and #6 (cumulative return compounding), which summed step returns instead of compounding portfolio values, causing significant metric errors over 252-step episodes. Bug #9 (adaptive TD error accumulation) caused TD error sums to grow unboundedly, eventually making all regimes appear equally important and collapsing the adaptive strategy to uniform sampling; this was fixed by introducing EMA decay ($\alpha = 0.99$). Bugs #1–#9 were fixed and all experiments re-executed from scratch.

Bug #10 was discovered post-execution: the `add()` method in the replay buffer still accumulated TD errors without EMA decay, partially undoing the Bug #9 fix. After correction, TD3 adaptive dropped from Sharpe +0.733 to +0.501. Bug #11 fixed a look-ahead bias in the risk parity baseline (`iloc[-60:]` → `iloc[:60]`), changing its Sharpe from +0.003 to −0.055. E1 and E7 adaptive configurations were re-run after these fixes.

H Data Description

Table 15 summarises the five ETFs used in this study. The universe spans US equities, government bonds, commodities, and cash equivalents, enabling the agent to construct diversified portfolios across market conditions.

Table 15: Asset universe.

Ticker	Name	Asset Class
SPY	SPDR S&P 500 ETF Trust	US Large-Cap Equity
QQQ	Invesco QQQ Trust (NASDAQ-100)	US Tech/Growth Equity
TLT	iShares 20+ Year Treasury Bond ETF	US Government Bonds
GLD	SPDR Gold Shares	Commodities (Gold)
BIL	SPDR Bloomberg 1–3 Month T-Bill ETF	Cash / Money Market

Daily OHLCV (Open, High, Low, Close, Adjusted Close, Volume) data was obtained from Yahoo Finance via the `yfinance` Python library, covering 2010-01-01 to 2023-12-31 (approximately 3,500 trading days). The data was split chronologically to prevent look-ahead bias: training on 2010–2019 (approximately 2,515 days), validation on 2020–2021 (approximately 503 days), and testing on 2022–2023 (approximately 499 days). Each evaluation episode spans one full year (252 trading days) with $1M initial capital.

The observation vector consists of 98 dimensions for off-policy algorithms (SAC, TD3, Rainbow) and 101 dimensions for PPO (which receives an additional 3-dimensional one-hot regime indicator). The base observation comprises 92 technical features plus 6 portfolio state variables (5 current weights and normalised portfolio value). Each asset receives 19 per-asset features, except SPY which receives 16 (relative strength is not computed for the benchmark asset against itself), giving a total of $16 + (4 \times 19) = 92$ market features. Table 16 lists all per-asset features.

Table 16: Per-asset feature vector (19 features per asset, 16 for SPY). Features marked with † are excluded for SPY as relative strength cannot be computed for the benchmark against itself.

#	**Feature**	**Category**	**Parameters**	**Normalisation**
1	Daily returns	Returns	–	None
2	RSI	Momentum	14-day window	Min-max [0, 1]
3	MACD line	Momentum	Fast 12 / Slow 26 / Sig 9	Z-score (60d)
4	MACD signal line	Momentum	As above	Z-score (60d)
5	MACD histogram	Momentum	As above	Z-score (60d)
6	Bollinger lower band	Volatility	20-day, 2 std	Z-score (60d)
7	Bollinger middle band	Volatility	20-day SMA	Z-score (60d)
8	Bollinger upper band	Volatility	20-day, 2 std	Z-score (60d)
9	Bollinger Band width	Volatility	20-day, 2 std	Z-score (60d)
10	Bollinger %B	Volatility	20-day, 2 std	None (bounded [0,1])
11	ATR	Volatility	14-day window	Z-score (60d)
12	On-Balance Volume (OBV)	Volume	Cumulative	Z-score (60d)
13	Rolling volatility	Volatility	5-day window	Z-score (60d)
14	Rolling volatility	Volatility	10-day window	Z-score (60d)
15	Rolling volatility	Volatility	20-day window	Z-score (60d)
16	Relative strength†	Relative	5-day vs SPY	Z-score (60d)
17	Relative strength†	Relative	10-day vs SPY	Z-score (60d)
18	Relative strength†	Relative	20-day vs SPY	Z-score (60d)
19	Bid-ask spread proxy	Liquidity	(High − Low) / Close	Z-score (60d)

Table 17 summarises the full observation vector composition.

Table 17: Observation vector composition by algorithm type.

Component	**Off-Policy (SAC, TD3, Rainbow)**	**On-Policy (PPO)**
SPY features (16)	16	16
QQQ features (19)	19	19
TLT features (19)	19	19
GLD features (19)	19	19
BIL features (19)	19	19
Current portfolio weights	5	5
Normalised portfolio value	1	1
One-hot regime indicator	–	3
Total	**98**	**101**

The reduced feature set used in E4 retains only the daily returns column per asset (feature 1), removing all technical indicators. This reduces the observation to 5 features (one per asset) plus 6 portfolio state variables, giving an 11-dimensional observation.

Regime labels are assigned using a rule-based detector by default: a bull regime is identified when the 50-day SMA exceeds the 200-day SMA and 20-day rolling volatility is below the 75th percentile of historical volatility; a bear regime is identified when the 50-day SMA is below the 200-day SMA and volatility is above the 75th percentile; all other periods are classified as sideways. During the test period, the regime distribution is approximately 38% bull (192 days), 37% bear (183 days), and 25% sideways (124 days). The training period (2010–2019) is dominated by bull markets, reflecting the longest US bull run in history, which creates the regime imbalance that motivates regime-aware replay strategies.